

## **PRINCIPLE OF CONSERVATION OF MASS: LAVOISIER'S LAW (1st LAW) – 20 min**

### **Time required**

20 minutes

### **Equipment**

- 1 × Precision electronic balance (not included)
- 1 × 200 ml wide-mouth flask
- 1 × Rubber stopper with a hole for the flask
- 1 × Plastic film sheet
- 1 × Dropper
- 1 × 50 ml beaker
- 1 × Spatula
- Zinc crystals (reagent not included)
- 36% hydrochloric acid in a bottle (reagent not included)

### **EXPERIMENT IMAGES**

**STEP 1:** Flask with zinc crystals closed with a perforated stopper and a dropper containing hydrochloric acid

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### **Objective of Experiment No. 1**

The objective of this experiment is to verify the law of conservation of mass (Lavoisier's law), which states:

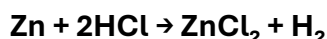
“In a chemical reaction, the sum of the masses of the reactants is equal to the sum of the masses of the products.”

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### **Description of the experiment**

- Using the spatula and the electronic balance, weigh and pour 1 g of zinc crystals into the flask.
- Pour hydrochloric acid into the beaker and then use the dropper to take an appropriate amount.
- Insert the dropper into the hole of the rubber stopper and close the flask (see STEP 1). To ensure maximum airtightness, plastic film can be placed around the stopper.
- Place the system obtained on the balance and record the total mass.

- At this point, remove the flask from the balance and press the dropper, releasing the hydrochloric acid onto the zinc, which will cause the following reaction to occur:



- What is produced is hydrogen in gaseous form, which remains inside the closed flask. By measuring again the mass of this new system, a value equal to the initial one is obtained, aside from experimental errors. This demonstrates that the mass of the reactants is equal to that of the products.

### Final Observation

At the end of the experiment, it was observed that the chemical transformation occurs without the dispersion of hydrogen ( $\text{H}_2$ ) and that the total mass of the system remains constant. It was precisely through this observation that Lavoisier's law of conservation of mass was demonstrated.

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### Questions for the teacher

1. Considering Lavoisier's law, think about a fireplace with burning wood. Is the mass of the resulting charcoal equal to the initial mass of the wood?
  2. Consider a beaker with two ice cubes of mass  $m = 50 \text{ g}$ . The beaker is heated so that the ice melts into water. Does this increase in temperature affect the final mass of the system? Can we still refer to Lavoisier's law?
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